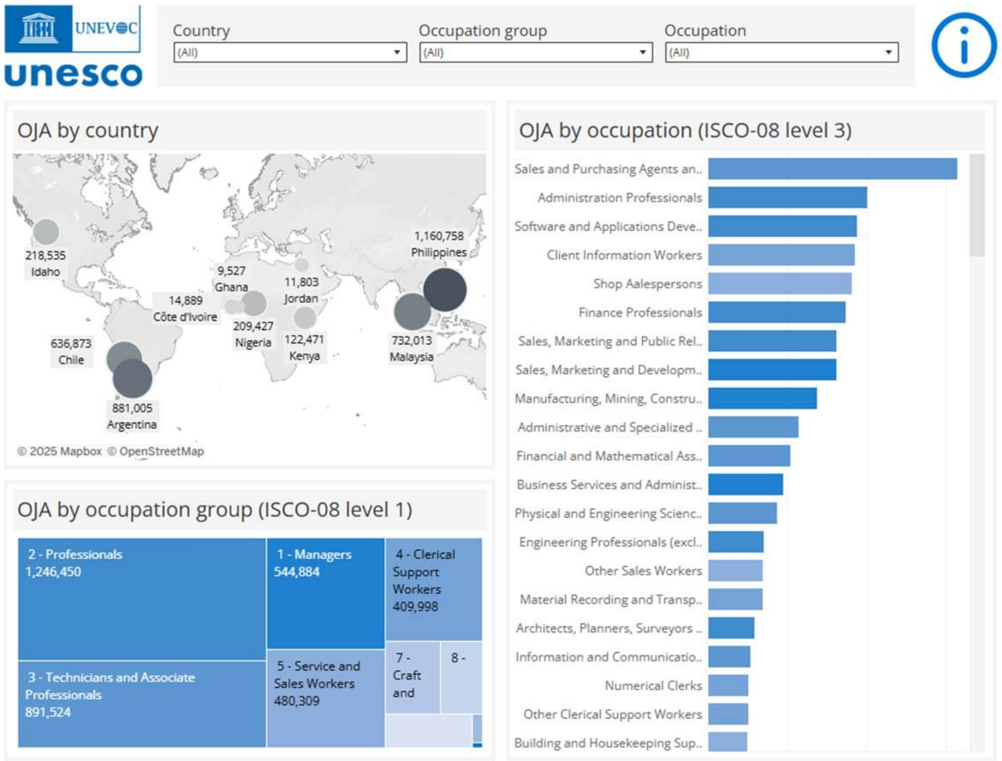
Global Skills Tracker: A tool for assessing skills demands

AI4EAC Innovation Challenge 2026 - 18 March

Global Skills Tracker

What is the Global Skills Tracker

The Global Skills Tracker is **a tool** that enables users to **explore skills data across** various dimensions, including **countries, industries and occupations**.



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What is the Global Skills Tracker

High-level Mapping

The tool will enable users to explore **skills data at a macro level**, covering different dimensions such as **countries, industries, and occupations**.

Trends

It includes **dynamic features** allowing users to **identify trends over time and interactive charts**.

Insights

By leveraging job posting analytics, the tool help users **identifying patterns in the demand for specific skills**, enabling a more **comprehensive understanding of the skills landscape**.

What is a online job postings?

Junior Analyst

• 3.5 ★

Zamfara

You must create an Indeed account before continuing to the company website to apply

Apply now

Job details

Job type

Full-time

Full job description

At , we believe in investing in our people to help them grow, both professionally and personally. The foundation of our business is strong relationship, with colleagues, clients and other stakeholders and we work hard every day to make this a reality. Our commitment is to create unlimited growth by giving our people continuous opportunities.

 global organisation extends across 167 countries and territories, with 88,120 people working out of 1,617 offices – and we're all working towards one goal: to provide our clients with exceptional service. Our firms across the organisation cooperate closely and comply with consistent operating principles and quality standards.

outh Africa has vacancies for Junior Analysts to join our Financial Services Technology Division in our Johannesburg and Cape Town Offices commencing in 2024.

Requirements

Qualifications and Experience:

- BSc Computer Science, BCom Information Systems / Informatics, BCom Internal Auditing or Equivalent
- Honours in any of the above is advantageous but not essential
- COBIT, ITIL, ISO27001-2 and other models is advantageous
- The Ideal candidate should be studying toward or wanting to study toward a Certified Information Systems Auditor (CISA) qualification

Competencies:

- Have a passion and/or interest in emerging technologies for example: machine learning, Internet of Things (IoT), Artificial Intelligence (AI) and robotics.
- Professionalism
- Strong client orientation
- Attention to detail
- Ability to priorities and handle stress
- Task driven & delivery focused
- Ability to work with all levels within an organization
- Socially aware and able to work as part of a diverse team

Core Competencies:

- Relationships and Collaboration
- Exceptional Client Service
- Business Growth
- Engaging people
- Leadership
- Quality, Risk management and Operational performance

Why a Global Skills Tracker (GST)

Growing interest in developing and implementing practical **labour market intelligence applications using internet-based labour market data** to support **evidence-based decision-making in policy design and evaluation**

Benefits of using **internet-based labour market data**

- More **frequent**
- More **timely**
- More **granular**
- **Less burden**
- **Cheaper**

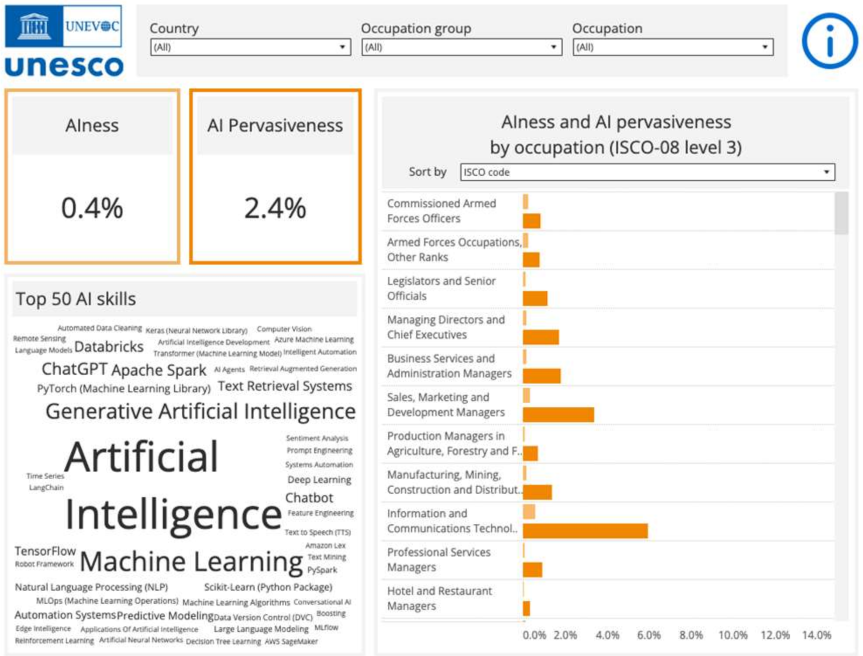
The ED/YLS section at UNESCO designed, developed and implemented a Global Skills Tracker to monitor skills demand, in alignment with our TVET Strategy 2022-2029.

Geographical coverage

The Global Skills Tracker will focus on the following countries:

- **Africa:** Kenya, Nigeria, Cote d'Ivoire, Ghana, Somalia, Morocco, Angola, Mozambique, Senegal, Gabon, Ethiopia, Togo, Tanzania, Uganda
- **Asia:** Malaysia, The Philippines
- **LAC:** Argentina, Chile
- **North America:** US (Idaho)
- **Middle East:** Jordan

Countries have been selected based on UNESCO's current projects, data availability and Lightcast expertise working with online job postings in those geographies.



The Global Skills Tracker is an example of **how AI technology can be leverage to the benefit of education and to anticipate skills demand** in this evolving sector.

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How can the Global Skills Tracker support our projects?

The use of the Global Skills Tracker can also help expand and diversify learning opportunities through higher technical and vocational education and training (TVET) and higher education

Share **real-time data on labour market conditions**

Dynamically **consolidate data on skills demand**

Guide **programme and policy development**

Monitor and analyse emerging skills within the rapidly evolving landscape of digital transformation

With a specific focus on key areas such as **digital, AI and green skills**, the tracker will provide valuable insights into the constantly evolving demands of the global workforce.

Levels of analysis

The Global Skills Tracker provides insights into the skills landscape across various dimensions such as countries, industries and occupations. It helps identify trends in skills demands, supports strategic decision-making in skills development and offers interactive dashboards for exploring skills data. The countries currently covered are Argentina, Chile, Côte d'Ivoire, Ghana, Idaho/USA, Jordan, Kenya, Malaysia, Nigeria and Philippines.

Labour market

Labour market indicators

Online job markets

Country analysis

Occupation analysis

Sector analysis

Skills insights

Green skills

Digital skills

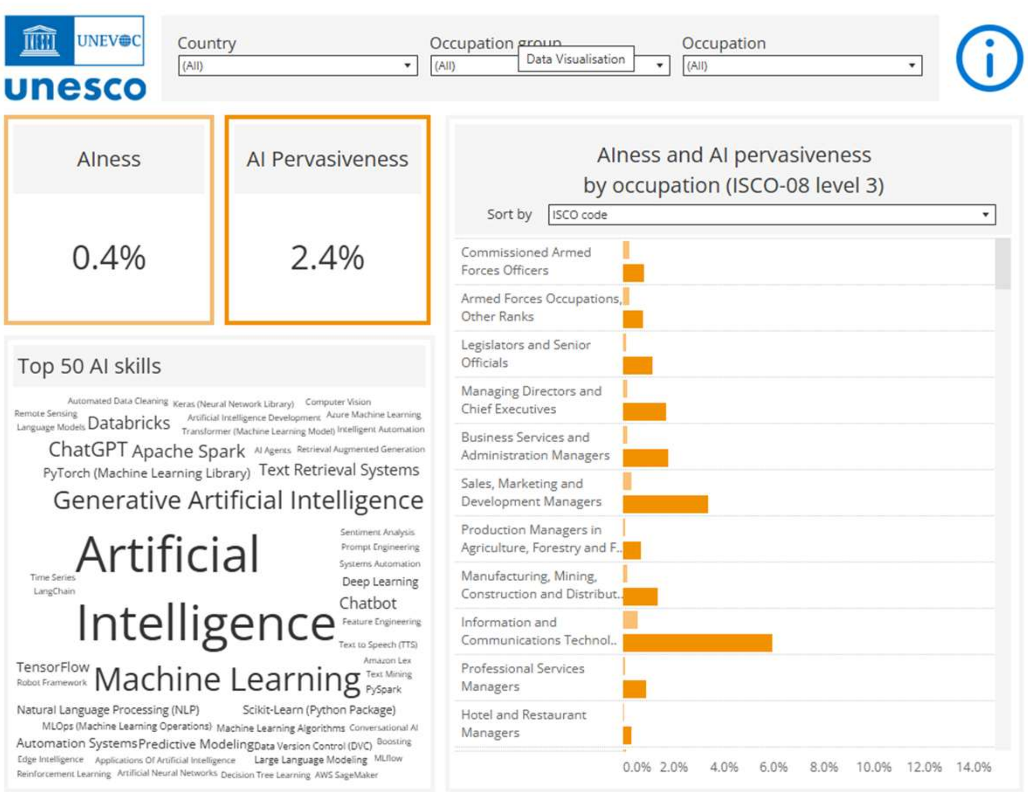
AI skills

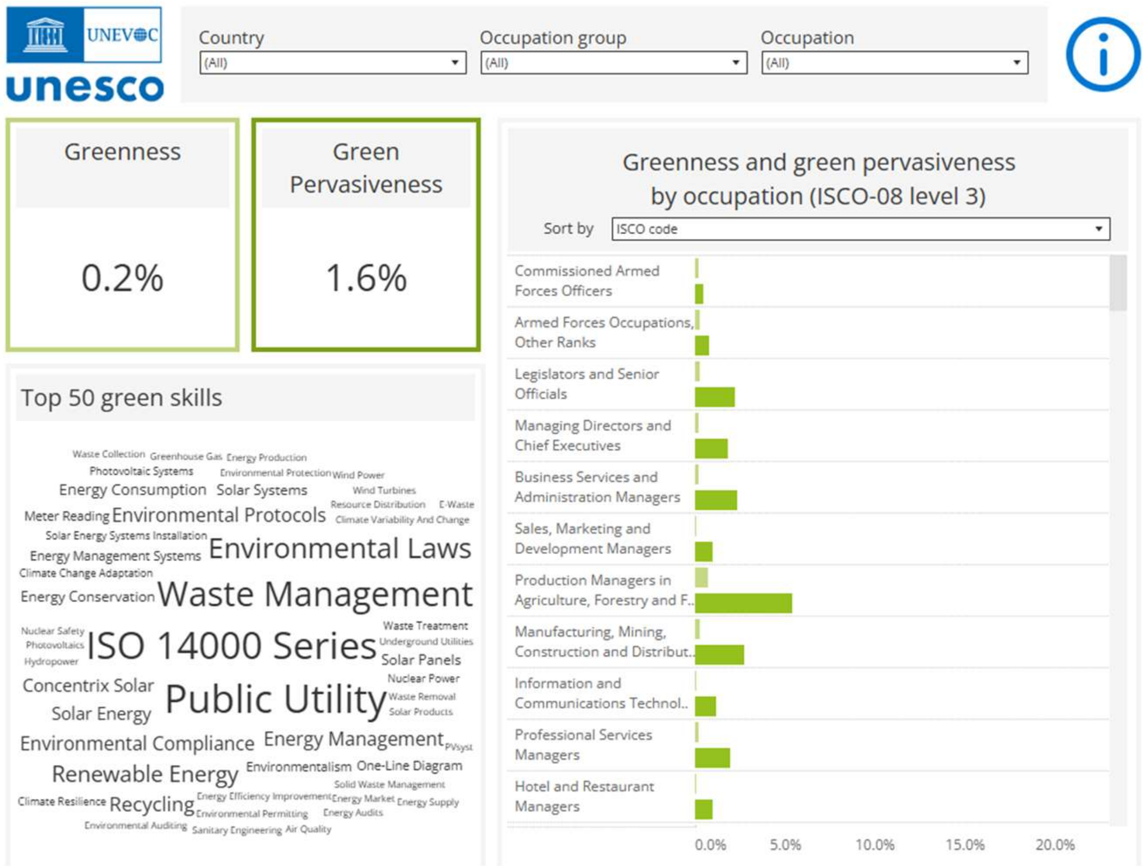
Compare data tool

Online job markets

Skills insights

Global Skills Tracker: 21 data sheets





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Understanding Green-Digital-AI Pervasiveness vs. Green-Digital-AIness Metrics

Greenness, Digitalness, AIness measure the intensity or concentration of a specific skill type.

They answer the question: Of all the skills mentioned in job advertisements for this occupation/sector, what proportion belongs to this category?

(Total mentions of green/digital/AI skills across all OJAs) / (Total mentions of all skills across all OJAs) × 100

Green-Digital-AI Pervasiveness metrics measure the spread or prevalence of a skill type across job advertisements.

They answer the question: How often do job advertisements for this occupation/sector mention at least one skill from this category?

(Number of OJAs containing at least one green/digital/AI skill) / (Total number of OJAs) × 100

Practical example

Scenario:
Data Scientist occupation - Analysis of 100 job advertisements
Skills observed across all OJAs:
- Total skills mentions: 1000
- Digital skill mentions: 600

OJAs mention at least one digital skill: 85

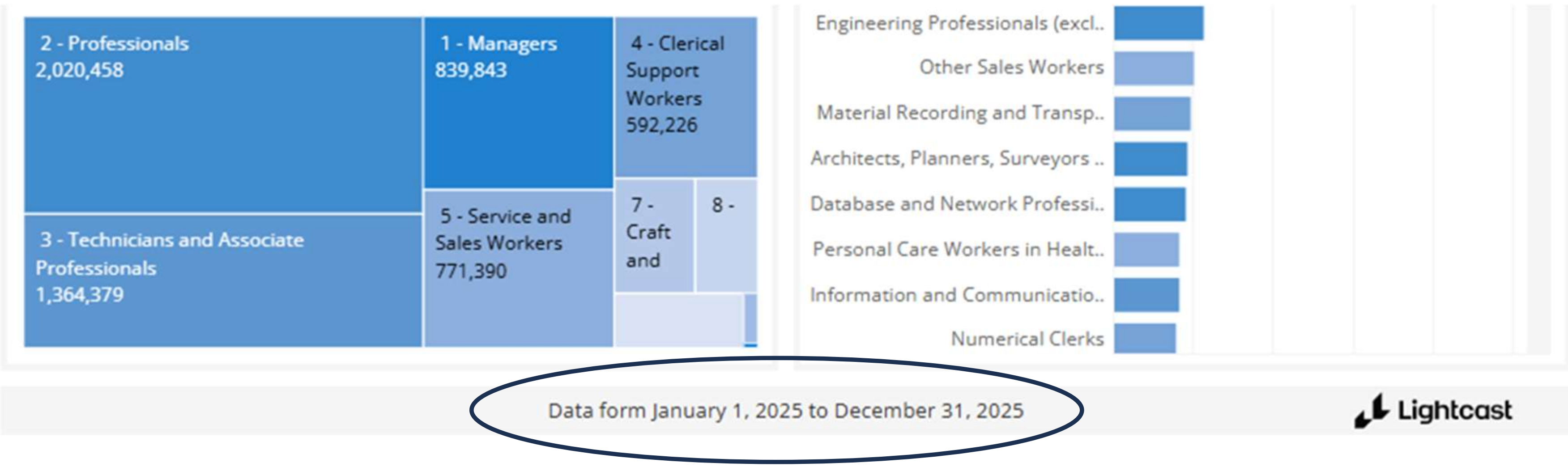
Results:
Digitalness = 600/1000 = 60%
digital skills represent 60% of all skill mentions

Digital Pervasiveness = 85/100 = 85%
85% of job advertisements require at least one digital skill

Analysis period

- The Global Skills Tracker is updated quarterly (every three months) and always reports data from the previous 12 months.
- Each update provides a rolling 12-month window of job advertisement data
- The specific time-period covered by the data is clearly indicated at the bottom of every dashboard.

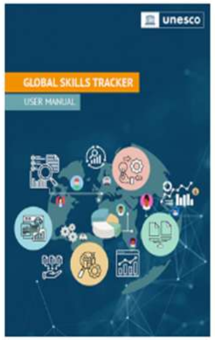
Q1 2026 Update (released February 2026)



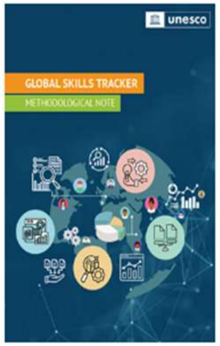
Support documents

Dashboard currently available in three languages: English, French, Portuguese

English

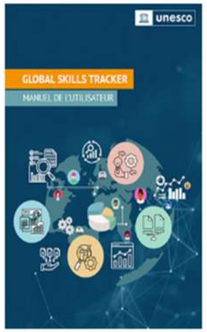


User manual

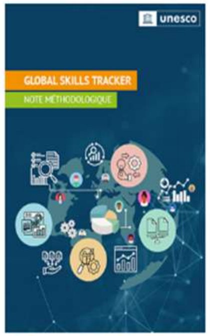


Methodological note

French

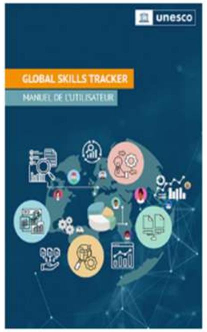


Manuel de l'utilisateur



Note méthodologique

Portuguese
(in development)



Assessment Guide, forthcoming

Taxonomies Used in the Global Skills Tracker (GST)

The Global Skills Tracker (GST) employs internationally recognized classification systems to ensure data consistency, comparability, and analytical rigor across countries and time periods.

Occupation ->ISCO (International Standard Classification of Occupations)
- Enables cross-country comparability of occupational data
- Provides a hierarchical structure (major groups, sub-major groups, minor groups, unit groups) that allows analysis at different levels of granularity
- Facilitates tracking of skill demand trends across specific occupations globally

Economic Activity-> ISIC (International Standard Industrial Classification of All Economic Activities)
- Allows analysis of skill demand by economic sector (e.g., manufacturing, services, construction)
- Supports policy-makers in identifying which sectors are driving demand for specific skill types
- Ensures international comparability of sectoral trends

Education -> ISCED (International Standard Classification of Education)
- Universally recognized framework adopted by countries worldwide
- Ensures comparability of educational data across different national education systems
- Applicable to diverse educational systems and contexts

Skills-> Lightcast Skills Taxonomy
- Provides a comprehensive and up-to-date classification of skills observed in real labor market data
- Distinguishes between specialized and common skills
- Organizes skills into hierarchical categories, enabling analysis of skill families and related competencies
- Based on millions of job advertisements, ensuring relevance to actual employer demand
- Allows precise identification and measurement of green, digital, and AI skills within job postings

How to download the visualization

To download the Global Skills Tracker or a specific chart:

- as an image -> Click the Download button in the bottom right toolbar, then select Image to save a snapshot of the current view.
- as a pdf file -> Click the Download button and select PDF. You can choose to export the current view or a specific chart.
- as a PowerPoint file -> Click the Download button and select PowerPoint to generate a presentation slide. You can choose to export the current view or a specific chart.

Access the Global Skills Tracker

SCAN ME

Exploring the tracker together:

- What do the top green skills can tell us about the sector?
- What would be your recommendation for someone interested in developing digital skills for the crafts sector in Malaysia?
- Share a surprising fact with us…

Explanation of the dataset provided

Global Skills Tracker

Asia images/Shutterstock.com

Goal of the challenge

Develop a method to predict the most relevant occupations starting from a small set of skills.

Input

- 5 skills
- Skills are extracted from the microdata behind the Global Skills Tracker

Using these skills, the goal is to: **Identify the five most similar occupations**

The challenge evaluates how well participants can:

- Use skill–occupation relationships
- Measure skill similarity / overlap
- Rank occupations based on relevance to the input skills

Expected Output

For each set of 5 skills:

- Top 5 predicted occupations
- Ranked by similarity / relevance

Training dataset

The training data is derived from the **microdata behind the Global Skills Tracker**, which links **job postings, skills and occupations**.

The dataset captures **real-world relationships between skills and occupations** observed in job postings. These relationships are used to train models that can **infer occupations from a set of skills**.

1. Job Postings

Each posting contains metadata such as: **Posting ID, Geography, Industry sector**

2. Posting–Skill Links

A second table links **job postings to the skills extracted from them**. For each posting we observe: **Skill ID, Skill category, Skill name**

3. Metadata

Each posting is associated with a **Lightcast Occupation Taxonomy (LOT) occupation**, including: *Occupation ID, Occupation name, Hierarchy information, Training requirements*

For each skill, you have a full reference to the **Lightcast Skills:** Skill ID, Skill name, Skill category, Skill description

Gold set

The **Gold Set is used to evaluate the performance of the models developed in the challenge**. Participants will train their models on the training data, but final evaluation is done on the Gold Set.

Each row (~4,800 examples) contains: **5 skills** *(skill_1 … skill_5, skill_name_1 … skill_name_5)*

Skill 1	Skill 2	Skill 3	Skill 4	Skill 5
Python	SQL	Data Analysis	Statistics	Machine Learning

For each skill set we provide the **5 most relevant occupations** *(occ_1 … occ_5, occ_name_1 … occ_name_5)*

These occupations represent the true occupations most associated with the given skills.

Each occupation also has **an associated score** *(score_1 … score_5)*

These scores indicate the strength of the relationship between the skill set and the occupation **(Higher scores = stronger relevance)**

Thank you

Learn more: www.unesco.org/education

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Youth, Literacy and Skills Development Section
(ED/PLS/YLS)
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About the Challenge

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Challenge Design

Skills2Job Intelligent Career Pathways Challenge

The task: Build a recommendation engine that rank the top 5 jobs based on 5 skills.

Data: Tabular

Multi-Metric Evaluation: map@5: Mean Average Precision @ 5

Submission Format requirements: ID columns + 5 columns, one per occupation

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Dataset Overview

You are given 3 datasets:

- Occupation information
- Skills information
- Job postings (real-world data)

[Skills (input)] → [Model] → [Top 5 Jobs (output)]

Test Set Structure

ID	skill_1	skill_2	skill_3	skill_4	skill_5
1	skill_1	skill_2	skill_3	skill_4	skill_5

each row contains 5 skills

Task Description

➔ For each row (5 skills)
➔ Predict the TOP 5 jobs

Expected Output

ID	occ_1	occ_2	occ_3	occ_4	occ_5
1	occ_1	occ_2	occ_3	occ_4	occ_5

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How Map@5 Works

MAP@5 scores based on how early correct jobs appear

The goal is to find correct jobs and place them as early as possible

Correct jobs ranked earlier receive a much higher score

Incorrect predictions significantly lower your overall MAP@5 score

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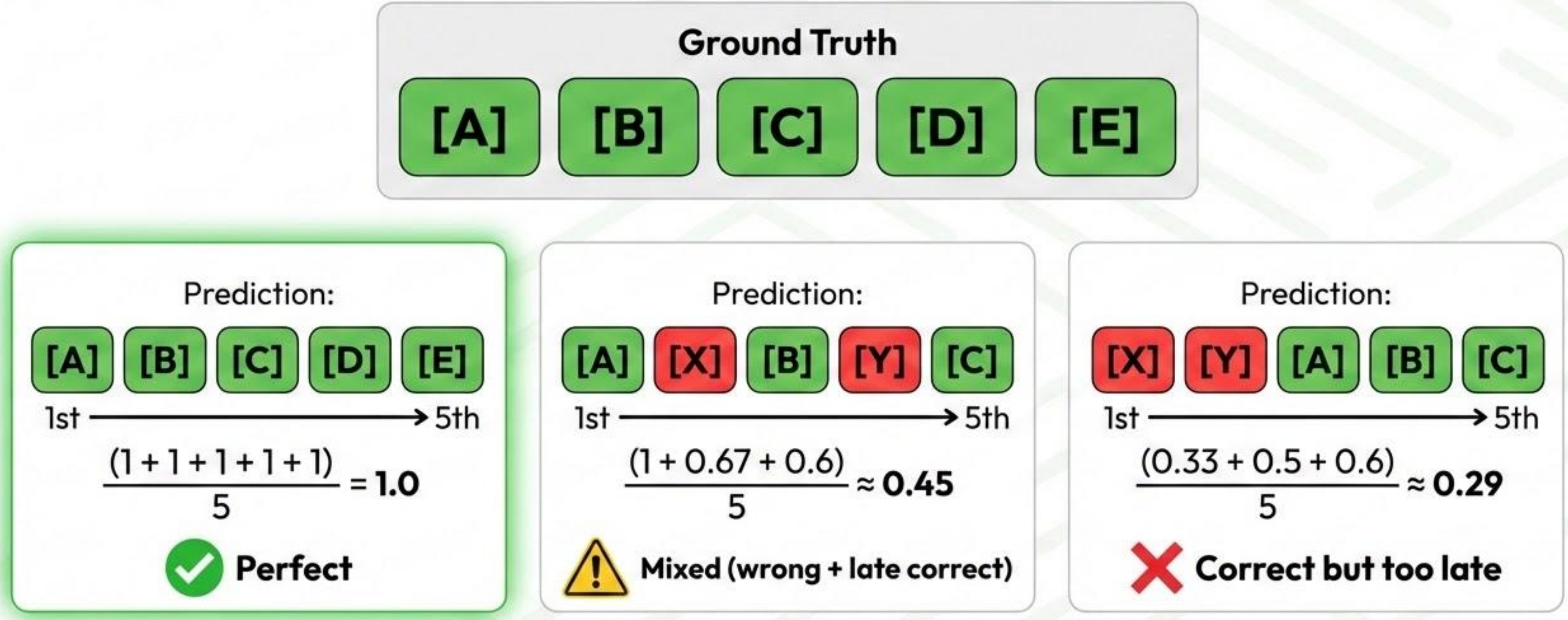
How Map@5 scores your predictions



Preparing for the Challenge

Practice Challenge: [AI4EAC Education Practice Challenge]

The task: predicting a child's ELOM score & Identify which early learning programme factors contribute to better learning outcomes in children.

Evaluation Metric:

- **RMSE: Root Mean Squared Error**

Key Preparation activities:

- Familiarise yourself with Zindi
- Understand the submission process
- Deal with ranking problems (even if you will not be evaluated on during this challenge)

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Key Skills You'll Develop

Working with Structured Data:

- Understand relationships between skills and jobs
- Combine multiple datasets effectively
- Extract useful features from real-world data

Machine Learning & Ranking:

- Build similarity-based recommendation systems
- Use embeddings or matching techniques
- Learn how to rank predictions (not just predict)

Mindset Shift:

→ From predicting individual outcomes
→ To ranking the most relevant results


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